

Mastering Loops in C

Comprehensive Guide to For Loops, Nesting, and Applications

1. For Loop in C

A **for loop** is an entry-controlled loop used when the number of iterations is known beforehand. It provides a concise way of writing the loop structure.

```
for(initialization; condition; update) {  
    // block of code to be executed  
}
```

Pro Tip: The initialization happens once, condition is checked before every iteration, and the update happens at the end of every iteration.

```
#include <stdio.h>  
int main() {  
    for(int i = 1; i ≤ 5; i++) {  
        printf("%d ", i);  
    }  
    return 0;  
}
```

2. Nested Loops in C

A nested loop is simply a loop inside another loop. The "inner loop" completes all its iterations for every single iteration of the "outer loop".

```
for(int i = 1; i ≤ 3; i++) {  
    for(int j = 1; j ≤ 2; j++) {  
        printf("i=%d j=%d\n", i, j);  
    }  
}
```

Common Uses for Nested Loops:

- **Pattern Printing:** Creating stars or number triangles.
- **Matrix Operations:** Traversing 2D arrays (Rows and Columns).
- **Multiplication Tables:** Generating tables for multiple numbers.

3. Real-Life Examples

Loops aren't just for math; they power everyday logic in software:

- **Education:** Printing marksheets for a class of 60 students.
- **E-commerce:** Calculating the total price of all items in a digital cart.
- **Daily Logs:** Tracking steps for 7 days of the week.

```
for(int day = 1; day ≤ 7; day++) {  
    printf("Day %d: Logged activity\n", day);  
}
```

4. Code Challenges (Practice)

Challenge 1: Basic Counting

Print numbers from 1 to 10 using a for loop.

```
for(int i = 1; i ≤ 10; i++) {  
    printf("%d ", i);  
}
```

Challenge 2: Logic Building

Try these on your own:

1. **Even Numbers:** Print all even numbers between 1 and 20.
2. **Multiplication Table:** Take an input 'N' and print its table up to 10.
3. **Star Pattern:** Print a 5x5 square of asterisks (*) using nested loops.

End of Comprehensive Loop Guide